

From Repair Correctness to Task Consequences

Completed local development experiment
and the remaining primary-study inputs

ACL 2027 research project

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Completed. The empirical pipeline now connects deletion-induced admissions to predictions and loss on separate diagnostic evaluation records. A fixed 112 deletion trajectories were evaluated at four checkpoints in two learner dimensions: **896 checkpoint evaluations**. Twelve selected checkpoints additionally passed strict exact-state and numerical-certificate audits. All execution occurred in this workspace; no synthetic empirical data or external compute was used.

Finding. In this small lexical preview, random deletion can admit previously excluded records, but the mean task effect is modest. Larger effects occur under topology stress. The initial curated model performs worse than simple reference models, so the preview does not establish useful semantic curation or a publication-ready NLP benefit.

Eight-deletion endpoint, 768-dimensional lexical learner. R samples all training records, U samples initially excluded records, and A is label-blind graph stress. Positive relative MSE effect means frozen selection has higher loss than the freshly recurated oracle.

Arm	Paths with additions	Mean additions	Effect (%)	Descriptive 95% interval
R	19/64	0.344	+0.100	[-0.043, +0.264]
U	17/32	0.906	+0.127	[-0.129, +0.450]
A	16/16	6.062	+4.453	[+4.274, +4.624]

The intervals condition on this one corpus, split and feature map. They do not include independent test-source uncertainty. Stress outcomes cannot be used as random-request prevalence, and no significance claim is made.

Study status. The main semantic study remains blocked on original source-rich corpus assets, pinned semantic embeddings or usable local encoder assets, and genuine human semantic-quality evidence. An executable intake gate and an exact file-format template now make the next import concrete.

What was fixed before outcomes

The same 100 Civil Comments preview records were grouped by NFC-normalized, whitespace-collapsed text, preserving case. A fixed salted hash assigned **82 development-training and 18 diagnostic-evaluation records**. There was no split retry. The records had already been available in earlier engineering work; the evaluation subset is not an untouched confirmatory test.

The curator uses only training text: 128-dimensional lexical word/bigram hashing, strict cosine threshold 0.6 and stable priority. It initially selects 55 of 82 training records and excludes 27, with 195 blocker edges. The learner uses either 64- or 768-dimensional lexical hashing, original fractional toxicity responses, $\lambda = 0.01$, no intercept and unbounded ridge scores. These are not E5 representations. Neither threshold nor regularization was tuned on these evaluation outcomes.

The runner saved the design, code/data hashes, partition IDs and graph-only requests before computing losses. Each dimension reuses the same 64 R, 32 U and 16 A trajectories at cumulative deletion counts 1, 2, 4 and 8. There is no source arm: this fixture lacks native article/source metadata.

Arm	Deleted	Paths with additions	Mean additions	Effect64 (%)	Effect768 (%)
R	1	2/64	0.031	+0.047	+0.010
R	2	5/64	0.078	-0.007	-0.007
R	4	10/64	0.156	+0.085	+0.012
R	8	19/64	0.344	+0.353	+0.100
U	1	7/32	0.219	-0.111	-0.069
U	2	12/32	0.406	+0.048	-0.100
U	4	15/32	0.625	+0.403	+0.079
U	8	17/32	0.906	+0.629	+0.127
A	1	0/16	0.000	+0.000	+0.000
A	2	0/16	0.000	+0.000	+0.000
A	4	16/16	2.000	+5.341	+2.262
A	8	16/16	6.062	+13.946	+4.453

All 658 zero-admission evaluation rows and all negative signed effects remain in the raw results. Tiny floating residues in zero-effect rows are retained in the ledger and round to zero here. The complete interval and distribution summaries for both dimensions accompany the report.

At checkpoints 1, 2 and 4, all 16 A trajectories have the same deletion set; their degenerate intervals do not demonstrate precision across distinct stress cases. At checkpoint 8, A has 15 distinct deletion sets; R and U have 64 and 32, respectively. Across all checkpoints there are 358 distinct deletion sets per dimension, not 448 distinct datasets.

No normalized-exact-text group crosses the partition. However, 119 training/evaluation lexical pairs exceed the threshold, and source/semantic leakage guards are incomplete. This is an evaluation-code diagnostic, not source-independent generalization evidence.

Reference models, numerical fidelity and interpretation

The oracle O-G independently rescores retained training records, selects them under the stated curator, rebuilds moments and solves ridge. The frozen-selection baseline B-F *refits* on surviving members of the initial selected set; it is not the unchanged predeletion model. Both use the same $(G + \lambda nI)W = H$ target. Evaluation records never enter either training state, the curation graph or request generation.

d	Curated MSE	Uncurated MSE	Same-size hash MSE	Mean baseline MSE
64	0.02694	0.01964	0.01708	0.02029
768	0.02364	0.02093	0.01919	0.02029

These are predeletion losses on the same 18 diagnostic records. The constant reference is the mean of the original training responses. The same-size control uses an outcome-blind ID hash. **Curated performance is worse in both dimensions.** This unfavorable result is preserved and blocks any utility success claim for the lexical operating point.

The reported relative effect is

$$\frac{\text{MSE}_{\text{frozen}} - \text{MSE}_{\text{oracle}}}{\text{MSE}_{\text{predelete}}}.$$

Raw prediction RMS, fixed-score-SD-normalized RMS, raw signed loss differences, admission frequency and unconditional/active-only admission counts are also saved. Zero denominators remain undefined rather than receiving epsilon floors. Within each arm/checkpoint, 10,000 bootstrap draws resample entire reset trajectories and retain paired method differences. Independently sampled arms are never paired by their ordinal names; checkpoints are not pooled.

Strict integration. At dimension 768, the first R, U and A paths were replayed through exact canonical state at all four checkpoints. All 12 repaired byte strings equalled fresh retained-data exact construction. All 12 head certificates met 10^{-10} tolerance; the largest certified radius was 4.5091×10^{-14} . This is a scoped certificate audit, not a claim that all 896 floating evaluation rows were certified.

Independent result audit. A separate verifier imports no production curation, repair or analysis module. It reconstructs partitions, requests, selected sets and metrics, and uses an independent dual ridge solve to check the runner’s primal solver. It checks source hashes, saved prediction arrays, bootstrap intervals, the utility controls and geometry. Its full count and error ledger is included. The largest independently observed prediction difference was below 1.14×10^{-15} .

Decision supported by this stage. The task-consequence measurement pipeline works on real labeled text and retains null/unfavorable outcomes. The next decisive experiment must use the planned semantic calibration and source-disjoint corpus panels. Increasing preview trajectory counts cannot replace those inputs.

Concrete next inputs and execution sequence

The first useful import is a **source-rich original Civil Comments bundle**. The planned route is TFDS CivilComments 1.2.4 with original article, publication, record and response fields. Provide either usable local pinned E5 encoder/tokenizer/runtime assets, or an aligned frozen FP32 semantic embedding cache with derivation evidence. No external job is required by the intake or repair pipeline.

1. **Inspect original assets.** Validate hashes, row identity, original labels and native source construction. A partial import can start preparation; it does not silently lower the main panel-size target.
2. **Prepare panels and semantic calibration.** Build source-disjoint training/calibration/test manifests and the prescribed cross-split guards. Create blinded pair assignments on calibration records only. Genuine human responses remain distinct from model-generated judgments.
3. **Lock the study.** Freeze encoder/tokenizer revisions, embedding cache, threshold and independent quality validation, regularization, source units, requests, code and resource envelope before confirmation.
4. **Run the planned evidence blocks.** Start the natural admission census and oracle/frozen-selection task effects, then compare exact/fast repair against the strongest eligible-payload baseline under matched access and cost accounting. Preserve infeasible and unfavorable cells. The second labeled corpus and full-refit boundary remain requirements of the complete program.

Executable intake. The included validator checks local file hashes, unique original IDs, source-unit derivation, source-disjoint partitions, exact-text/parent-link guards, label copying, finite aligned FP32 arrays and calibration response structure. Eleven intake software checks passed; the placeholder manifest correctly fails 17 mechanical checks. A mechanically consistent file does not authenticate human work or historical provenance. The validator therefore never automatically declares the entire study ready.

The cache interface is a rank-two FP32 `.npy` array aligned to a JSON row-ID list, with corpus and encoder/derivation manifests. Detailed requirements and a fillable JSON template are in the companion archive. Do not invent missing source IDs, labels, embeddings, approval evidence or human responses.

Replay and artifacts. The package includes the unchanged Civil fixture, the executed runner, independent audit, strict arithmetic backend, all predictions/results, intake files, protocol reference and integrity hashes. No CC-News article text is redistributed. Preserve the result directory before rerunning. From the archive root:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
export MKL_NUM_THREADS=1
python3 empirical_execution/phase2/run_development.py
```

The original main-study design remains unchanged; the new development addendum documents the stronger exact mode and its distinct numerical target. Earlier engineering records remain available separately. No background jobs continue after this report.

Official acquisition references: [TensorFlow Datasets Civil Comments](#); [multilingual E5 model card](#).